

Useful Memory under Drift

Temporal-Validity Horizons for Finite-Memory Distribution Tracking

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Abstract

We study the temporal validity of retained evidence for distribution tracking under drift. The central object is a temporal-validity horizon: the memory scale at which the gain from larger samples is overtaken by drift-induced staleness. A general law balances a finite-sample term and a staleness term, yielding an optimal memory scale and a useful-memory region whose normalized geometry depends only on the finite-sample and drift exponents. We prove this law in a tractable one-dimensional root- n proof model, show through a Gaussian lower-bound construction and a Gaussian location benchmark that the horizon is structural, and extend the horizon law to finite projection families and one-dimensional fixed- ε Sinkhorn geometry. An inferable-horizon horizon-inference analysis propagates lag-geometry uncertainty into data-driven memory selection through an identifiability criterion and a quadratic regret law. Experiments reveal interior U-curves, roughness-dependent scaling, a real-stream finite-memory signature, and a regime in which temporal validity fails before standard changepoint alarms.

Keywords: tracking under drift; temporal validity; finite-memory tracking; distribution tracking; Wasserstein distance; Sinkhorn divergence.

MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Retained evidence under drift is simultaneously informative and obsolete. Larger memory reduces finite-sample noise, but it also increases the age of the observations used to estimate the present. Finite-memory distribution tracking therefore has a temporal-validity horizon: a largest memory scale that remains useful before drift-induced staleness overtakes the remaining sampling gain.

This horizon is governed by the balance law

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample rate exponent of the chosen geometry, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Balancing the finite-sample term against the staleness term yields the temporal-validity horizon

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The same envelope induces a useful-memory region: the memory lengths whose tracking error stays within a fixed factor of the optimum. Its normalized geometry depends only on (a, H) .

This object is structural, but operational use requires more than the existence of an optimizer. Memory selection from data depends on whether noisy lagged transport discrepancies can localize the horizon within the useful-memory region. That question leads to horizon inference and to an identifiability criterion for useful-memory selection.

The theorem line closes in a tractable one-dimensional bounded-support fixed-span triangular-array model. There, the quantile representation of W_2^2 , a triangular-array Bahadur expansion, and a design-effect comparison yield a root- n finite-sample term under explicit compact-support, fixed-span, and integrated-remainder hypotheses. A structural Gaussian lower bound shows that the horizon is imposed by drift geometry rather than by a particular estimator, and the Gaussian location model supplies a matching benchmark on deterministic Hölder paths. Beyond that proof model, finite projection families and sliced Wasserstein yield transfer results under uniform projected control, while in fixed- ε Sinkhorn geometry one-dimensional inheritance is explicit on compact support and the embedded low-dimensional extension is reduced to a conditional theorem problem. Empirically, the same object appears through interior U-curves, roughness-dependent scaling, real-stream diagnostics, and validity loss before the first detector alarm.

Five contributions:

- (1) **Temporal-validity horizon and useful-memory region.** The temporal-validity horizon governs finite-memory distribution tracking under drift, together with a useful-memory region whose normalized geometry is determined by (a, H) .
- (2) **Closed theorem line in a tractable proof model.** A one-dimensional root- n triangular-array proof model, the exact uniform-window staleness bound, a structural Gaussian lower bound, and a Gaussian location benchmark show that the horizon is structural.
- (3) **Transfer and operational geometry.** Finite projection families and sliced geometries yield transfer theorems, one-dimensional fixed- ε Sinkhorn inheritance is proved, and the embedded low-dimensional Sinkhorn step is isolated as a conditional theorem problem.

- (4) **Horizon inference and identifiability.** Plug-in horizon inference yields an identifiability criterion for useful-memory selection and a quadratic regret law for horizon mislocation.
- (5) **Empirical signatures of temporal validity.** Empirical results exhibit interior U-curves, roughness-dependent scaling, real-stream finite-memory diagnostics, and pre-detection validity loss.

Memory length under drift is a statistical scale set jointly by finite-sample geometry and temporal roughness.

2 Useful Memory and Temporal-Validity Horizons

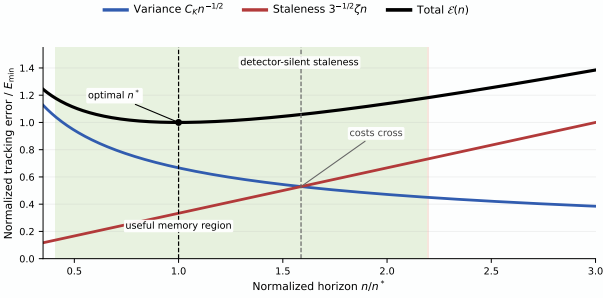


Figure 1: Finite-sample-staleness trade-off under drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the temporal-validity horizon n^* . The green band marks a near-optimal useful-memory region, and the red band marks pre-detection validity loss: retained evidence can stop representing the present before enough changepoint evidence accumulates to trigger an alarm.

Throughout, P_t denotes the present target law, $\bar{P}_t^{(n)}$ the window target obtained by averaging the last n target laws, and $\hat{P}_t^{(n)}$ the estimator built from the last n observations. The finite-sample term always refers to the estimation error relative to $\bar{P}_t^{(n)}$, while the staleness term refers to the discrepancy between $\bar{P}_t^{(n)}$ and P_t .

Finite-sample error relative to $\bar{P}_t^{(n)}$ and staleness relative to P_t combine into a horizon law. When both sides are controlled, the resulting scale induces a useful-memory region: the range of memory lengths whose tracking error remains near optimal.

2.1 General Horizon Law

The main theorem separates finite-sample behavior from drift-induced staleness.

Theorem 2.1 (Abstract upper law). *Let d be a probability metric satisfying the triangle inequality. At time t , let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\hat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E} d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C_K n^{-a}$$

and

$$d(\bar{P}_t^{(n)}, P_t) \leq C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d(\hat{P}_t^{(n)}, P_t) \leq d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) + d(\bar{P}_t^{(n)}, P_t).$$

Taking expectations and applying the two assumptions yields the claim.

The theorem isolates the structure of the problem: the path class supplies the staleness term, while the chosen geometry and sample regime supply the finite-sample term.

Proposition 2.2 (Triangular-array transfer for Hadamard-differentiable functionals). *Let $\mathcal{T}$ be a functional on probability laws that is Hadamard differentiable at P_t tangentially to a linear subspace of signed measures, with derivative $\dot{\mathcal{T}}_{P_t}$ bounded on the relevant tangent class. Suppose that for the triangular-array estimator and its window target, the base metric satisfies*

$$\mathbb{E} d(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C_K n^{-a}$$

and the drift envelope satisfies

$$d(\bar{P}_t^{(n)}, P_t) \leq C_S \zeta n^H.$$

If the remainder in the Hadamard expansion is $o(n^{-a})$ uniformly on the relevant class, then there exist constants $\tilde{C}_K, \tilde{C}_S > 0$ such that

$$\mathbb{E} \|\mathcal{T}(\hat{P}_t^{(n)}) - \mathcal{T}(P_t)\| \leq \tilde{C}_K n^{-a} + \tilde{C}_S \zeta n^H.$$

Proof. Write the Hadamard expansion at P_t and apply the boundedness of the derivative to the centered triangular-array fluctuation. The drift term contributes through the base-metric envelope because $\mathcal{T}$ is locally Lipschitz on the class. The remainder is lower order by assumption, so the transformed law inherits the horizon envelope.

Proposition 2.3 (Finite-dimensional smooth transfer). *Let $\phi_1, \dots, \phi_k$ be bounded measurable functions and let $Z(P) = (\mathbb{E}_P[\phi_1], \dots, \mathbb{E}_P[\phi_k])$. Let $g \in C^2(U)$ on an open set $U \subset \mathbb{R}^k$ containing the relevant range of Z . Define $T(P) = g(Z(P))$. Suppose that on the triangular-array window target and estimator,*

$$\mathbb{E} \|Z(\hat{P}_t^{(n)}) - Z(\bar{P}_t^{(n)})\| \leq C_K n^{-a}$$

and

$$\|Z(\bar{P}_t^{(n)}) - Z(P_t)\| \leq C_S \zeta n^H.$$

If the Hessian of g is bounded by M on the relevant range, then

$$\mathbb{E} |T(\hat{P}_t^{(n)}) - T(P_t)| \leq \tilde{C}_K n^{-a} + \tilde{C}_S \zeta n^H,$$

and the Taylor remainder is quadratic in the moment error:

$$\begin{aligned} |T(P') - T(P) - \nabla g(Z(P))^\top (Z(P') - Z(P))| \\ \leq \frac{M}{2} \|Z(P') - Z(P)\|^2. \end{aligned}$$

Proof. Apply the multivariate Taylor formula to g around $Z(P_t)$. The bounded Hessian gives the stated quadratic remainder. The first-order term is controlled by the assumed finite-sample and drift envelopes for Z , and the argument from proposition 2.2 yields the horizon law for T .

Corollary 2.4 (Temporal-validity horizon). *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a , H , and C_S , at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimal error scale.

Write the continuous optimizer of Φ as

$$n_\star := \left(\frac{aC_K}{HC_S\zeta} \right)^{1/(a+H)}.$$

This optimizer differs from the scale $n^*(a, H)$ only by constants depending on a , H , and C_S , so the asymptotic horizon law remains unchanged.

2.2 Useful-Memory Region

The horizon scale induces a near-optimal region of memory lengths. For any $\delta > 0$, define the useful-memory region by

$$\mathcal{U}_\delta := \{n > 0 : \Phi(n) \leq (1 + \delta)\Phi(n_\star)\}.$$

This object records how much memory remains acceptable up to a fixed tolerance.

Proposition 2.5 (Useful-memory region at tolerance δ). *Let $\Phi(n) = C_K n^{-a} + C_S \zeta n^H$ with $a, H > 0$, and let $n_\star$ be its continuous optimizer. Then the normalized profile*

$$\Psi(x) := \frac{\Phi(n_\star x)}{\Phi(n_\star)} = \frac{Hx^{-a} + ax^H}{a + H}$$

is strictly decreasing on $(0, 1]$ and strictly increasing on $[1, \infty)$. Consequently,

$$\mathcal{U}_\delta = [n_-, n_+] = n_\star [x_-(\delta; a, H), x_+(\delta; a, H)],$$

where $x_-(\delta; a, H) \leq 1 \leq x_+(\delta; a, H)$ are the two solutions of $\Psi(x) = 1 + \delta$. In particular, the useful-memory region scales linearly with the horizon scale $n_\star$, and its relative shape depends only on a , H , and δ .

Proof. Since $n_\star$ minimizes Φ , differentiation gives

$$aC_K n_\star^{-a} = HC_S \zeta n_\star^H.$$

Substituting $n = n_\star x$ into Φ and dividing by $\Phi(n_\star)$ yields the displayed identity for Ψ . Differentiation then gives

$$\Psi'(x) = \frac{aH}{a + H} (x^{H-1} - x^{-a-1}),$$

which is negative for $x < 1$ and positive for $x > 1$. Since $\Psi(1) = 1$ and $\Psi(x) \rightarrow \infty$ as $x \rightarrow 0$ or $x \rightarrow \infty$, the sublevel set $\{x > 0 : \Psi(x) \leq 1 + \delta\}$ is a compact interval containing 1, which proves the representation of $\mathcal{U}_\delta$.

For small tolerance, the useful-memory region has the local expansion

$$x_\pm(\delta; a, H) = 1 \pm \sqrt{\frac{2\delta}{aH}} + o(\sqrt{\delta}), \quad \delta \downarrow 0,$$

because $\Psi''(1) = aH$. The local curvature therefore controls how quickly the near-optimal region contracts around the horizon.

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.3 Drift Regularity and Staleness Accumulation

The staleness side of the law is controlled by a deterministic path class. For $H \in (0, 1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \{(P_s)_{s \in \mathbb{Z}} : W_2(P_{t-j}, P_t) \leq \zeta j^H \text{ for all } j \geq 0\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.6 (Hölder staleness bound via averaged couplings). *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2(\bar{P}_t^{(n)}, P_t) \leq \frac{\zeta^2}{n} \sum_{j=0}^{n-1} j^{2H},$$

and therefore, by Jensen's inequality,

$$W_2(\bar{P}_t^{(n)}, P_t) \leq \zeta \left(\frac{1}{n} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2} = C_{H,n} \zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)} \sum_{j=0}^{n-1} j^{2H} \right)^{1/2}.$$

In particular, $C_{H,n} \rightarrow (2H + 1)^{-1/2}$ as $n \rightarrow \infty$.

Proof. For each lag j , let π_j be an optimal coupling between P_{t-j} and P_t . Their average

$$\bar{\pi} := \frac{1}{n} \sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t , so

$$\begin{aligned} W_2^2(\bar{P}_t^{(n)}, P_t) &\leq \int \|x - y\|^2 d\bar{\pi}(x, y) \\ &= \frac{1}{n} \sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t). \end{aligned}$$

Applying the Hölder envelope gives the displayed finite- n constant, and taking square roots completes the proof.

Combining lemma 2.6 with corollary 2.4 yields the horizon family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root- n finite-sample regime $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite- n constant in lemma 2.6 is the constant used here for uniform windows; the simpler $(2H + 1)^{-1/2}$ expression is its large-window limit.

2.4 One-Dimensional Proof Model for the Root- n Regime

The bounded-support fixed-span one-dimensional triangular array provides a tractable rigorous finite-sample instantiation of the abstract law. The model is built on common compact support, uniformly regular window mixtures, and an integrated Bahadur remainder hypothesis.

Proposition 2.7 (Root- n finite-sample bound in the 1D proof model). *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^n P_{n,j}.$$

Assume:

- (i) *all $P_{n,j}$ are supported on a common compact interval $[L, U]$;*
- (ii) *the within-window drift span is uniformly bounded in n ;*
- (iii) *each mixture law $\bar{P}_n$ has a density $\bar{f}_n$ on $[L, U]$ that is bounded below by a positive constant and uniformly Hölder;*
- (iv) *writing q_n for the quantile function of $\bar{P}_n$ and $\hat{q}_n$ for the empirical quantile function of $\hat{P}_n^{\text{tri}}$, one has the Bahadur representation*

$$\begin{aligned} \hat{q}_n(u) - q_n(u) &= \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u), \\ &\quad u \in (0, 1), \end{aligned}$$

with

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

Then

$$\mathbb{E} W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1}),$$

and therefore, by Jensen's inequality,

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))}.$$

Then

$$\mathbb{E} \int_0^1 (\hat{q}_n(u) - q_n(u))^2 du = \mathbb{E} \int_0^1 (Z_n(u) + R_n(u))^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\hat{F}_n(x) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{X_{n,j} \leq x\},$$

independence across the row gives

$$\text{Var}(\hat{F}_n(q_n(u))) = \frac{1}{n^2} \sum_{j=1}^n p_{n,j}(u)(1 - p_{n,j}(u)) \leq \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E} \int_0^1 Z_n(u)^2 du \leq m^{-2} \int_0^1 \text{Var}(\hat{F}_n(q_n(u))) du \leq \frac{1}{4m^2n}.$$

By item (iv),

$$\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is then $o(n^{-1})$ by Cauchy-Schwarz, since $\mathbb{E} \int_0^1 Z_n(u)^2 du = O(n^{-1})$ and $\mathbb{E} \int_0^1 R_n(u)^2 du = o(n^{-1})$. Hence the expected squared Wasserstein error is $O(n^{-1})$, and Jensen's inequality gives the root- n rate for $\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n)$.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root- n .

2.5 Verifiable Sufficient Conditions for the 1D Proof Model

An explicit deformation class makes the proof-model hypotheses verifiable. Let

$$g_\delta(x) = x + \delta x(1 - x), \quad x \in [0, 1], \quad |\delta| \leq \Delta < 1.$$

Then g_δ is an increasing C^2 diffeomorphism of $[0, 1]$, with

$$1 - \Delta \leq g'_\delta(x) \leq 1 + \Delta, \quad |g''_\delta(x)| \leq 2\Delta,$$

so the transformed densities satisfy

$$\frac{1}{1+\Delta} \leq f_\delta(y) \leq \frac{1}{1-\Delta},$$

and the same support is preserved.

Corollary 2.8 (Verifiable Bahadur remainder for the smooth deformation class). *Assume that each row law has the form*

$$P_{n,j} = (g_{\delta_{n,j}})_\# \text{Unif}[0, 1], \quad |\delta_{n,j}| \leq \Delta < 1,$$

and that the within-window span is uniformly bounded in n . Then item (iv) of proposition 2.7 holds. More precisely, there exists a constant $C_\Delta < \infty$ such that

$$\mathbb{E} \int_0^1 R_n(u)^2 du \leq C_\Delta n^{-2}.$$

Consequently,

$$\mathbb{E} W_2(\hat{P}_n^{\text{tri}}, \bar{P}_n) = O(n^{-1/2})$$

on this deformation class.

Proof. Write $\bar{F}_n$ and q_n for the mixture cdf and quantile. Since each $g_{\delta_{n,j}}$ is an increasing diffeomorphism of $[0, 1]$, every mixture law $\bar{P}_n$ is supported on $[0, 1]$ and has density $\bar{f}_n = \frac{1}{n} \sum_{j=1}^n f_{\delta_{n,j}}$ satisfying the explicit bounds

$$\frac{1}{1+\Delta} \leq \bar{f}_n(y) \leq \frac{1}{1-\Delta}, \quad \|\bar{f}_n'\|_\infty \leq \frac{2\Delta}{(1-\Delta)^3}.$$

Hence the inverse map $\bar{F}_n^{-1}$ is uniformly twice differentiable on $(0, 1)$, with second derivative bounded by a constant depending only on Δ . Applying the second-order Taylor expansion of $\bar{F}_n^{-1}$ around $u = \bar{F}_n(q_n(u))$ gives

$$\hat{q}_n(u) - q_n(u) = \frac{u - \hat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u),$$

with

$$|R_n(u)| \leq C_\Delta |\hat{F}_n(q_n(u)) - u|^2.$$

Now

$$\hat{F}_n(q_n(u)) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{X_{n,j} \leq q_n(u)\}$$

is an average of independent Bernoulli variables with mean u . Uniformly in u , its fourth centered moment is $O(n^{-2})$. Therefore

$$\mathbb{E} R_n(u)^2 \leq C_\Delta \mathbb{E} |\hat{F}_n(q_n(u)) - u|^4 \leq C'_\Delta n^{-2},$$

and integrating over $u \in (0, 1)$ yields

$$\mathbb{E} \int_0^1 R_n(u)^2 du \leq C'_\Delta n^{-2} = o(n^{-1}).$$

The root- n claim then follows from proposition 2.7. The same smoothness package is consistent with the classical Bahadur theory of Bahadur [1966] and van der Vaart [1998, Chapter 21], but here the explicit deformation bounds make the integrated remainder quantitative.

Remark 2.9 (Weak rowwise dependence). The proof of proposition 2.7 uses independence within a row only through the $O(n^{-1})$ variance bound for the empirical cdf term and through item (iv). Consequently, the same root- n conclusion persists under any short-range dependent row-wise model for which the empirical cdf still has L_2 fluctuation of order $n^{-1/2}$ and the integrated Bahadur remainder remains $o(n^{-1})$. Short-range mixing conditions are one natural route to such a package, but no general dependent-row theorem.

On the tested smooth fixed-support family, the integrated residual decays at empirical rates between 1.40 and 1.54 for spans 0.2 and 0.4, and the scaled quantity $n \int_0^1 R_n(u)^2 du$ still decays at rates between 0.40 and 0.54; see appendix B. On the fixed-span triangular-array uniform-kernel model, the residual-to-MSE ratio drops from about 0.38 at $n = 25$ to about 0.11 at $n = 400$, with integrated residual rates 1.43 for the triangular sample and 1.49 for the i.i.d. mixture benchmark. These diagnostics do not replace a general verification theorem, but they sit behind an explicit sufficient-condition corollary on the smooth deformation class and quantify span dependence on the tested classes.

2.6 Structural Lower Bound in the Root- n Regime

The upper law would be substantially weaker if it were only an estimator-specific statement. The lower bound shows that, in the root- n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

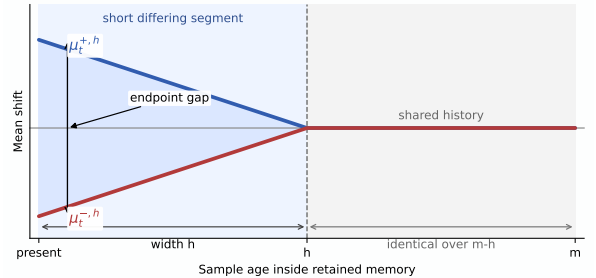


Figure 2: Gaussian lower-bound construction. The two paths share most of the retained history and differ only on a recent segment of width h , yet remain separated at the present. That recent segment already produces the same horizon exponent as the upper law in the root- n regime.

Proposition 2.10 (Structural lower bound in the root- n regime). *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm, h} = \pm \beta(h-j)_+^H, \quad j = 0, 1, \dots, h,$$

where $\beta \leq \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H, \zeta}^{\text{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm, h}, \sigma^2)$. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\hat{P}} \sup_{P \in \mathcal{G}_{H, \zeta}^{\text{Gauss}}} \mathbb{E} W_2(\hat{P}_t, P_t) \geq c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, in the root- n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof. Compare two hypotheses generated by the path pair $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H . On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^h r^{2H} / \sigma^2.$$

Choosing

$$\beta = \min \left\{ \zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^h r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H . Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law. The exact Gaussian two-point calculation used here is recorded in appendix A.

The lower bound is subclass based and not class-tight, but it is enough to show that the horizon is a structural feature of the problem and not an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes the constant-level geometry. The following proposition gives the sharp Gaussian constants available within that proof scheme.

2.7 Gaussian Lower-Bound Geometry

Let Φ and φ denote the standard normal distribution function and density.

Proposition 2.11 (Gaussian lower-bound constants). *Fix $H \in (0, 1]$ and set $p_H := 2H/(2H+1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H} \Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

For the Gaussian ramp profile $\mu_{t-j}^{\pm,h} = \pm \zeta(h-j)_+^H$, let

$$L_h^{\text{ramp}}(\sigma, \zeta) := \zeta h^H \Phi \left(-\frac{\zeta}{\sigma} \sqrt{\sum_{r=1}^h r^{2H}} \right).$$

Then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 (1 - (1-u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

If

$$L_h^{\min}(\sigma, \zeta) := \zeta h^H \Phi \left(-\frac{\zeta}{\sigma} \sqrt{\sum_{r=1}^h (g_r^{\min})^2} \right),$$

then, as $\sigma/\zeta \rightarrow \infty$,

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For $H < 1$, this strictly improves the ramp constant by the factor

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{H+1}{2H^2} \right)^{H/(2H+1)} > 1,$$

while at $H = 1$ the two profiles coincide.

Proof. The exact Gaussian testing problem reduces the two-point lower bound to the displayed quantities L_h^{ramp} and $L_h^{\min}$. After rescaling by $h = A(\sigma/\zeta)^{2/(2H+1)}$, the ramp energy and endpoint-minimal energy have limits $(2H+1)^{-1}$ and I_H , respectively. The corresponding normalized payoff becomes a scalar multiple of $x^{p_H} \Phi(-x)$, whose unique maximizer is x_H . The endpoint-minimal profile minimizes testing energy among all endpoint-saturating Hölder profiles, so it yields the sharper constant. Full details are given in appendix A.

These constants sharpen the lower-bound picture without changing the horizon exponents or the status of the main results.

The same lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Theorem 2.12 (Gaussian location minimax benchmark on the deterministic Hölder class). *Fix $H \in (0, 1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s \in \mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \leq \zeta j^H \quad \text{for all } j \geq 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \quad j = 0, 1, \dots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimal finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma, \zeta) := \inf_{n \geq 1} \inf_{\hat{P}_t^{(n)}} \sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t).$$

Then there exist constants $0 < c_H \leq C_H < \infty$, depending only on H , such that

$$\begin{aligned} c_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} &\leq \mathcal{R}_H^*(\sigma, \zeta) \\ &\leq C_H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}. \end{aligned}$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$Hy_H(2\Phi(y_H) - 1) = \varphi(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\text{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\text{mean}} = ((H+1)y_H)^{-1/(2H+1)} \times \left(2\varphi(y_H) + y_H(2\Phi(y_H) - 1) \right).$$

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\hat{\mu}_t^{(n)} := \frac{1}{n} \sum_{j=0}^{n-1} Y_{t-j}$$

and set $\hat{P}_t^{(n)} = N(\hat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2(N(m_1, \sigma^2), N(m_2, \sigma^2)) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. The noise term is $\sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\leq \frac{\zeta}{n} \sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu \in \mathfrak{M}_H(\zeta)} \mathbb{E} W_2(\hat{P}_t^{(n)}, P_t) \leq \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n} \sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H , and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm, h} = \pm \beta(h^H - j^H)_+, \quad j \geq 0,$$

with $\beta \leq \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm, h} - \mu_{t-j}^{\pm, h}| = \beta \min(h^H, j^H) \leq \zeta j^H.$$

Moreover, for fixed endpoint value βh^H , it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Minimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem.

2.8 Multivariate Extension via Finite Projection Families, Sliced, and Product Geometries

The horizon law also extends beyond one dimension when the multivariate geometry reduces to a uniformly controlled family of one-dimensional transport problems. This is the natural setting for sliced or coordinate-averaged Wasserstein constructions, where the finite-sample term is built by integrating projected one-dimensional errors.

Corollary 2.13 (Finite-family projection transfer). *Let $\Pi = \{\theta_1, \dots, \theta_m\}$ be a finite family of linear maps from the ambient space to $\mathbb{R}$, and define*

$$d_\Pi(P, Q) := \left(\frac{1}{m} \sum_{r=1}^m W_2^2(P_{\theta_r}, Q_{\theta_r}) \right)^{1/2}.$$

Assume that for every $r = 1, \dots, m$:

- (i) the projected triangular-array window satisfies the hypotheses of proposition 2.7 with constants uniform in r ;
- (ii) the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in r .

Then there exist constants $C_{K, \Pi}, C_{S, \Pi} < \infty$ such that

$$\mathbb{E} d_\Pi(\hat{P}_t^{(n)}, P_t) \leq C_{K, \Pi} n^{-1/2} + C_{S, \Pi} \zeta n^H.$$

Consequently,

$$n_\Pi^*(H) \asymp (C_{K, \Pi}/\zeta)^{1/(H+1/2)}.$$

In particular, the same law holds for coordinate-averaged product constructions obtained by taking Π to be the coordinate projections.

Proof. For each r , proposition 2.7 gives

$$\mathbb{E} W_2^2(\hat{P}_{t, \theta_r}^{(n)}, \bar{P}_{t, \theta_r}^{(n)}) \leq C n^{-1}$$

with C uniform in r . Averaging over r yields

$$\mathbb{E} d_\Pi^2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C n^{-1},$$

and Jensen gives the root- n bound for $\mathbb{E} d_\Pi(\hat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound averages in the same way:

$$d_\Pi(\bar{P}_t^{(n)}, P_t) \leq C_{S, \Pi} \zeta n^H.$$

Applying theorem 2.1 with $d = d_\Pi$ yields the displayed envelope and horizon scale.

Corollary 2.14 (Sliced-Wasserstein transfer under uniform projected control). *Let ρ be a probability measure on a family of one-dimensional projections θ , and define*

$$SW_{2, \rho}^2(P, Q) := \int W_2^2(P_\theta, Q_\theta) \rho(d\theta),$$

where P_θ denotes the pushforward of P under projection θ . Assume that for every θ in the support of ρ :

- (i) the projected triangular-array window satisfies the hypotheses of proposition 2.7 with constants uniform in θ ;
- (ii) the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in θ .

Then there exist constants $C_K, C_S < \infty$ such that

$$\mathbb{E} SW_{2,\rho}(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-1/2} + C_S \zeta n^H.$$

Consequently,

$$n^*(H) \asymp (C_K/\zeta)^{1/(H+1/2)}.$$

Finite projection families are included by taking ρ supported on finitely many atoms; corollary 2.13 states that case directly for multivariate tracking.

Proof. For each projection θ , proposition 2.7 gives

$$\mathbb{E} W_2^2(\hat{P}_{t,\theta}^{(n)}, \bar{P}_{t,\theta}^{(n)}) \leq C n^{-1}$$

with C uniform in θ . Integrating over ρ yields

$$\mathbb{E} SW_{2,\rho}^2(\hat{P}_t^{(n)}, \bar{P}_t^{(n)}) \leq C n^{-1},$$

and Jensen gives the root- n bound for $\mathbb{E} SW_{2,\rho}(\hat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound integrates in the same way, giving

$$SW_{2,\rho}(\hat{P}_t^{(n)}, P_t) \leq C_S \zeta n^H.$$

Applying theorem 2.1 with $d = SW_{2,\rho}$ yields the stated envelope and horizon scale.

2.9 Weighted-Memory Summaries

The main line above is stated for uniform windows, but the same decomposition extends to non-uniform lag weights. For a finite-memory estimator with weights $w_0, \dots, w_{n-1}$, define the effective sample size

$$n_{\text{eff}}(w) := \left(\sum_{j=0}^{n-1} w_j^2 \right)^{-1}$$

and the weighted lag moment

$$M_H(w) := \sum_{j=0}^{n-1} j^H w_j.$$

Any weighted horizon law of the form

$$\mathbb{E} d(\hat{P}_t^{(w)}, P_t) \leq C_K n_{\text{eff}}(w)^{-\alpha} + C_S \zeta M_H(w)$$

balances the finite-sample and staleness terms through these two summaries.

Proposition 2.15 (Uniform weights maximize effective sample size). *Let $w_0, \dots, w_{n-1}$ be nonnegative weights with $\sum_{j=0}^{n-1} w_j = 1$. Then*

$$n_{\text{eff}}(w) \leq n,$$

with equality if and only if $w_j = 1/n$ for all j . Moreover, if the drift profile is an exact linear ramp in the sense that the lag- j contribution is proportional to j , then the weighted staleness equals the effective lag.

Proof. By Cauchy–Schwarz,

$$1 = \left(\sum_{j=0}^{n-1} w_j \right)^2 \leq n \sum_{j=0}^{n-1} w_j^2,$$

so $n_{\text{eff}}(w) = 1/\sum_j w_j^2 \leq n$, with equality exactly when the weights are all equal. On an exact linear ramp, averaging the lag contribution with weights w_j recovers the weighted mean lag.

2.10 Benchmark Theorems and Operational Extensions

The Gaussian benchmark closes a full path-class model, while the fixed- ε Sinkhorn results place the same horizon logic in an operational transport geometry. The linear-Gaussian entropic-OT setting suggests an intermediate recursive regime between the i.i.d. fixed- ε theory and the embedded low-dimensional extension [Kalman, 1960, Jazwinski, 1970, Akyildiz et al., 2024].

Fixed- ε Sinkhorn setting. For fixed- ε debiased Sinkhorn, the central question is whether a practical geometry can supply a stable finite-sample term on the embedded fixed-span model. The target law is

$$\mathbb{E} S_\varepsilon(\hat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}) \leq C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed- ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark.

Definition 2.16 (Fixed- ε Sinkhorn setting). Fix a bounded embedded fixed-span model, an intrinsic complexity index k , a dual Hölder smoothness level α , and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We say that this model is in the fixed- ε Sinkhorn setting if:

- (i) the i.i.d. fixed- ε benchmark on the same support class admits a finite-sample bound with exponent a_ε on $\mathcal{B}$;
- (ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;
- (iii) the dual class lies in the parametric adaptation region $\alpha > k/2$.

Proposition 2.17 (Structural ingredients for fixed- ε Sinkhorn). *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.16 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$.*

Proposition 2.18 (Conditional embedded moderate-band Sinkhorn inheritance). *Fix the calibrated embedded fixed-span model with intrinsic dimension $k = 2$ and a compact moderate band $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}]$ bounded away from zero. Assume:*

- (i) a centered self-coupling stability lemma holds uniformly on $\mathcal{B}$ with a spectral-radius bound strictly below one and a uniform inverse-norm bound on the centered subspace;

- (ii) the centered Sinkhorn scaling map admits a uniform implicit-function linearization on $\mathcal{B}$ with remainder $o(n^{-1/2})$;
- (iii) the induced influence class satisfies a root- n triangular empirical- process bound on the embedded support class;
- (iv) the embedded fixed-span support and dual-complexity hypotheses of proposition 2.17 hold.

Then there exists a finite constant $C_{\mathcal{B}}$ such that

$$\sup_{\varepsilon \in \mathcal{B}} \mathbb{E} S_{\varepsilon} \left(\hat{P}_{t, \text{tri}}^{(n)}, \bar{P}_t^{(n)} \right) \leq C_{\mathcal{B}} n^{-1/2}.$$

Consequently the corresponding temporal-validity horizon on S_{ε} has the root- n horizon scale

$$n_{\mathcal{B}}^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Under (i)–(iii), the regularized Sinkhorn value admits a uniform first-order expansion on $\mathcal{B}$ with a remainder lower order than $n^{-1/2}$. The embedded fixed-span support and dual-complexity hypotheses in (iv) supply the same structural comparison needed in the 1D proof model. Applying the abstract horizon law with $a = 1/2$ yields the stated envelope and the induced horizon scale.

Proposition 2.19 (Pointwise fixed- ε inheritance in the 1D proof model). *Assume the hypotheses of proposition 2.7, let $\hat{P}_n^{\text{iid}}$ be the empirical measure of n i.i.d. samples from $\bar{P}_n$, and fix $\varepsilon > 0$. For the debiased Sinkhorn divergence S_{ε} with squared-Euclidean cost on the common compact support, one has*

$$\mathbb{E} S_{\varepsilon} \left(\hat{P}_n^{\text{tri}}, \bar{P}_n \right) = O(n^{-1/2}).$$

Proof. Write $V_{\varepsilon}(\mu, \nu)$ for the regularized OT value so that

$$S_{\varepsilon}(\mu, \nu) = V_{\varepsilon}(\mu, \nu) - \frac{1}{2} V_{\varepsilon}(\mu, \mu) - \frac{1}{2} V_{\varepsilon}(\nu, \nu).$$

Because all measures are supported on one compact interval, they lie in a common bounded quadratic-moment class. By Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4], for fixed ε there exists $L_{\varepsilon} < \infty$ depending only on that class and on the cost such that

$$|V_{\varepsilon}(\mu, \nu) - V_{\varepsilon}(\mu', \nu')| \leq L_{\varepsilon} (W_2(\mu, \mu') + W_2(\nu, \nu')).$$

Applying this bound to the cross term and both self terms gives

$$|S_{\varepsilon}(\mu, \nu) - S_{\varepsilon}(\mu', \nu')| \leq 2L_{\varepsilon} (W_2(\mu, \mu') + W_2(\nu, \nu')).$$

With $\nu = \nu' = \bar{P}_n$,

$$S_{\varepsilon} \left(\hat{P}_n^{\text{tri}}, \bar{P}_n \right) \leq S_{\varepsilon} \left(\hat{P}_n^{\text{iid}}, \bar{P}_n \right) + 2L_{\varepsilon} W_2 \left(\hat{P}_n^{\text{tri}}, \hat{P}_n^{\text{iid}} \right).$$

Taking expectations and using the triangle inequality gives

$$\mathbb{E} W_2 \left(\hat{P}_n^{\text{tri}}, \hat{P}_n^{\text{iid}} \right) \leq \mathbb{E} W_2 \left(\hat{P}_n^{\text{tri}}, \bar{P}_n \right) + \mathbb{E} W_2 \left(\hat{P}_n^{\text{iid}}, \bar{P}_n \right).$$

The first term is $O(n^{-1/2})$ by proposition 2.7. The second is the classical i.i.d. one-dimensional quantile-process rate

on the same compact support class [Bahadur, 1966, van der Vaart, 1998], so it is also $O(n^{-1/2})$. Hence

$$\mathbb{E} W_2 \left(\hat{P}_n^{\text{tri}}, \hat{P}_n^{\text{iid}} \right) = O(n^{-1/2}).$$

Finally, under the null comparison with fixed ε and compact support, the i.i.d. debiased Sinkhorn term satisfies

$$\mathbb{E} S_{\varepsilon} \left(\hat{P}_n^{\text{iid}}, \bar{P}_n \right) = O(n^{-1})$$

by del Barrio et al. [2023, Thm. 5.1, after fixed- ε rescaling]; see also the null-limit law in Goldfeld et al. [2024]. Combining the bounds yields the displayed $O(n^{-1/2})$ rate.

Proposition 2.20 (Bandwise fixed- ε inheritance in the 1D proof model). *Let $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0, \infty)$ be a compact regularization band. Under the hypotheses of proposition 2.19 on a common compact support class, there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon \in \mathcal{B}} \mathbb{E} S_{\varepsilon} \left(\hat{P}_n^{\text{tri}}, \bar{P}_n \right) \leq C_{\mathcal{B}} n^{-1/2}.$$

Consequently, the induced horizon scale is uniform on $\mathcal{B}$:

$$\sup_{\varepsilon \in \mathcal{B}} n_{\varepsilon}^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Repeat the proof of proposition 2.19. The stability constant L_{ε} from Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4] is bounded on $\mathcal{B}$ because all measures live in one compact support class and $\varepsilon \geq \varepsilon_{\min} > 0$. The i.i.d. debiased-Sinkhorn benchmark on compact support is also uniform on $\mathcal{B}$ by the fixed- ε asymptotics of Goldfeld et al. [2024] and the compact-support null bounds of del Barrio et al. [2023, Thm. 5.1, after fixed- ε rescaling]. Since the triangular-to-i.i.d. W_2 comparison is independent of ε , the constants in the pointwise proof can be chosen uniformly over $\mathcal{B}$.

Proposition 2.21 (Conditional embedded fixed- ε band inheritance). *Assume the fixed- ε Sinkhorn setting from definition 2.16. Suppose that on a compact band $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}]$, the centered Sinkhorn scaling map admits a uniform linearization with remainder $o(n^{-1/2})$ and a uniform self-coupling inverse bound. Then there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon \in \mathcal{B}} \mathbb{E} S_{\varepsilon} \left(\hat{P}_{t, \text{tri}}^{(n)}, \bar{P}_t^{(n)} \right) \leq C_{\mathcal{B}} n^{-1/2}.$$

Consequently, on any such band the induced horizon scale satisfies

$$\sup_{\varepsilon \in \mathcal{B}} n_{\varepsilon}^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. The linearization hypothesis reduces the triangular-array Sinkhorn divergence to its first-order influence term plus an $o(n^{-1/2})$ remainder. The uniform self-coupling inverse bound keeps the influence operator bounded across $\mathcal{B}$, while the triangular-array fluctuation of the empirical measure is root- n on the embedded fixed-span class. The same balance then gives the uniform $n^{-1/2}$ bound and the stated horizon scale.

Conjecture 2.22 (Bandwise fixed- ε Sinkhorn horizon inheritance). *Beyond the one-dimensional fixed- ε band result above, assume the fixed- ε Sinkhorn setting from definition 2.16. Then, on bands where the i.i.d. benchmark stays in the parametric Sinkhorn regime, the triangular-array finite-sample term should satisfy*

$$\mathbb{E} S_\varepsilon(\hat{P}_{t,\text{tri}}^{(n)}, \bar{P}_t^{(n)}) \leq C_\varepsilon n^{-1/2}, \quad \varepsilon \in \mathcal{B},$$

and the corresponding horizon scale should satisfy

$$n_\varepsilon^*(H) \asymp (C_\varepsilon/\zeta)^{1/(H+1/2)}.$$

More generally, if the i.i.d. benchmark on $\mathcal{B}$ has exponent a_ε , then the operational horizon law should be

$$n_\varepsilon^*(a_\varepsilon, H) \asymp (C_\varepsilon/\zeta)^{1/(a_\varepsilon+H)}.$$

Evidence. The i.i.d. benchmark comes from existing fixed- ε entropic transport results, including fixed-marginal limit theory for the Sinkhorn divergence and intrinsic-complexity adaptation [Goldfeld et al., 2024, Stromme, 2023, Groppe and Hundrieser, 2024]. In the embedded fixed-span model considered here, support-complexity inheritance is exact and the dual smoothness threshold reduces to $\alpha > k/2$. Calibration then asks whether triangular and i.i.d. rate estimates remain close across the regularization band. Additional one-dimensional smooth fixed-support diagnostics show that triangular and i.i.d. fitted exponents can agree closely on moderate fixed- ε bands, while direct Jacobian diagnostics concentrate the remaining difficulty on uniform control of the self-coupling derivative blocks across nontrivial bands and higher-dimensional embedded classes.

Regular dominated families remain the natural family-level extension class. Support-changing and singular families fall outside the present regime.

Once the finite-sample regime and the roughness level are fixed, the law yields the memory scale and the corresponding error scale. Rougher drift and weaker finite-sample rates imply shorter memory and larger residual error.

3 Horizon Inference and Identifiability

The horizon law is structural, but operational use requires more than the existence of an optimizer. A data-driven memory rule needs estimators of the drift-geometry parameters (H, ζ) together with a criterion that compares inferential resolution to the curvature of the useful-memory region. This section turns temporal validity into an inferable object.

Write $\alpha = \log \zeta$, and let m denote the per-lag sample size used to estimate lagged transport discrepancies. For a fixed lag design $1, \dots, L$, define the lag-energy information scale

$$\mathcal{I}_{m,L}(H) := m \sum_{j=1}^L j^{2H}.$$

Assumption 3.1 (Lag-geometry inference input). For the lag model $D_j = \zeta j^H$, suppose there are estimators $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ such that, on compact parameter sets,

$$\sqrt{\mathcal{I}_{m,L}(H)} \begin{pmatrix} \hat{\alpha}_{m,L} - \alpha \\ \hat{H}_{m,L} - H \end{pmatrix} \Rightarrow \mathcal{N}(\mathbf{0}, \Sigma_{L,H,\zeta}).$$

The power-law lag-geometry model of Parreira [2026b] provides one concrete source of this assumption.

The continuous optimizer of the horizon envelope can be written as the smooth map

$$n_*(\alpha, H) = \left(\frac{aC_K}{H C_S e^\alpha} \right)^{1/(a+H)},$$

$$g(\alpha, H) := \log n_*(\alpha, H).$$

Its gradient is

$$\nabla g(\alpha, H) = \begin{pmatrix} -(a+H)^{-1} \\ -[H(a+H)]^{-1} - g(\alpha, H)(a+H)^{-1} \end{pmatrix}.$$

Theorem 3.2 (Plug-in Horizon Central Limit Law). *Under Assumption 3.1,*

$$\sqrt{\mathcal{I}_{m,L}(H)} (\log \hat{n}_* - \log n_*) \Rightarrow \mathcal{N}(0, \tau^2),$$

where $\hat{n}_* = n_*(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ and

$$\tau^2 = \nabla g(\alpha, H)^\top \Sigma_{L,H,\zeta} \nabla g(\alpha, H).$$

Proof. The map g is smooth on any compact set bounded away from $H = 0$ and $\zeta = 0$. Apply the multivariate delta method to $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ under Assumption 3.1.

The useful-memory region is the set

$$\mathcal{U}_\delta = \{n : \Phi(n) \leq (1 + \delta)\Phi(n_*)\}.$$

Let $(x_-(\delta; a, H), x_+(\delta; a, H))$ be the normalized endpoints from Proposition 2.5, and define the log-radius

$$r_\delta(a, H) := \min\{-\log x_-(\delta; a, H), \log x_+(\delta; a, H)\}.$$

This quantity is the largest symmetric log-error around n_* that stays inside the useful-memory region.

Theorem 3.3 (Operational Identifiability of Useful Memory). *Under Assumption 3.1,*

$$\mathbb{P}(\hat{n}_* \in \mathcal{U}_\delta) = 2\Phi_N\left(\frac{\sqrt{\mathcal{I}_{m,L}(H)} r_\delta(a, H)}{\tau}\right) - 1 + o(1),$$

where Φ_N is the standard normal CDF. In particular,

$$\frac{\sqrt{\mathcal{I}_{m,L}(H)} r_\delta(a, H)}{\tau(H, \zeta)} \rightarrow \infty \implies \mathbb{P}(\hat{n}_* \in \mathcal{U}_\delta) \rightarrow 1.$$

Proof. By Proposition 2.5, $\hat{n}_* \in \mathcal{U}_\delta$ if and only if $|\log \hat{n}_* - \log n_*| \leq r_\delta(a, H)$. Standardize this event with Theorem 3.2 and use the symmetry of the Gaussian limit.

The quantity

$$\mathfrak{S}_{\delta,m,L} := \frac{\sqrt{\mathcal{I}_{m,L}(H)} r_\delta(a, H)}{\tau(H, \zeta)}$$

is therefore the operational identifiability score of the horizon. It compares lag-geometry information to the curvature-imposed precision requirement of the useful-memory region.

Proposition 3.4 (Deterministic stability of the horizon map). *Let $K \subset \mathbb{R} \times (0, \infty)$ be compact. Then there exists a finite constant L_K such that, for all $(\alpha_1, H_1), (\alpha_2, H_2) \in K$,*

$$|g(\alpha_1, H_1) - g(\alpha_2, H_2)| \leq L_K(|\alpha_1 - \alpha_2| + |H_1 - H_2|).$$

Consequently, if (α, H) and $(\hat{\alpha}_{m,L}, \hat{H}_{m,L})$ lie in K and

$$|\hat{\alpha}_{m,L} - \alpha| + |\hat{H}_{m,L} - H| \leq \frac{r_\delta(a, H)}{L_K},$$

then $\hat{n}_\star \in \mathcal{U}_\delta$.

Proof. The gradient of g is continuous on K , so $\|\nabla g\|_1$ is bounded there. Apply the mean-value theorem to obtain the Lipschitz bound. The inclusion claim follows from $|\log \hat{n}_\star - \log n_\star| \leq r_\delta(a, H)$ together with the characterization of $\mathcal{U}_\delta$ from Proposition 2.5.

The informative regime lies across the transition from low to moderate identifiability. Table 1 shows representative points across that transition.

Table 1: Representative regimes across the inferable-horizon transition.

L	m	H	δ	$\mathfrak{S}$	Empirical	Gaussian
5	20	0.3	0.050	0.501	0.411	0.384
8	10	0.9	0.020	1.001	0.633	0.683
3	200	0.3	0.010	1.996	0.956	0.954
8	100	0.3	0.005	3.001	0.996	0.997

On the tested grid, the plug-in central limit law is also well calibrated at the scale level: the empirical standard deviation of $\log \hat{n}_\star$ tracks the delta-method prediction closely, and the nominal 95% Gaussian interval coverage stays near its target across the reported regimes. The simulation suite also calibrates the regret expansion directly.

Proposition 3.5 (Deterministic and asymptotic plug-in horizon regret). *Let $u = \log(\hat{n}_\star/n_\star)$. Then, as $u \rightarrow 0$,*

$$\frac{\Phi(\hat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} = \Psi(e^u) - 1 = \frac{He^{-au} + ae^{Hu}}{a + H} - 1.$$

Consequently, for every $r > 0$,

$$|u| \leq r \implies \frac{\Phi(\hat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} \leq B_r(a, H),$$

where

$$B_r(a, H) := \max\{\Psi(e^r) - 1, \Psi(e^{-r}) - 1\}.$$

Moreover,

$$\frac{\Phi(\hat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} = \frac{aH}{2}u^2 + O(u^3).$$

Consequently,

$$\mathbb{E} \left[\frac{\Phi(\hat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} \right] = \frac{aH}{2} \frac{\tau^2}{\mathcal{I}_{m,L}(H)} + o(\mathcal{I}_{m,L}(H)^{-1}).$$

Proof. By Proposition 2.5,

$$\frac{\Phi(n_\star e^u)}{\Phi(n_\star)} = \Psi(e^u) = \frac{He^{-au} + ae^{Hu}}{a + H}.$$

The deterministic bound follows by maximizing this exact expression over $u \in [-r, r]$. Differentiate at $u = 0$. The first derivative vanishes and the second derivative equals aH , giving the quadratic expansion. The expectation bound then follows from Theorem 3.2 and the boundedness of third moments on compact parameter sets.

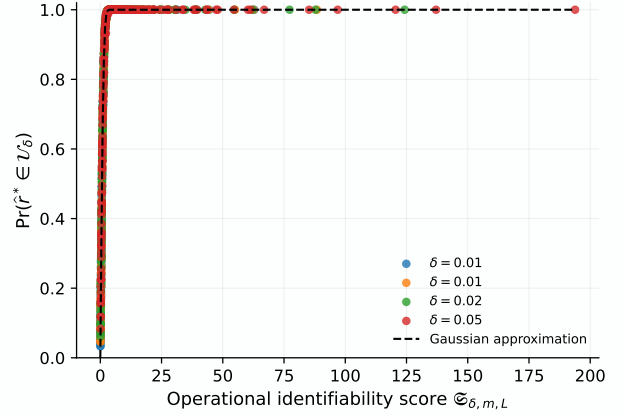


Figure 3: Operational identifiability of the useful-memory region. Across the tested grid, useful-region hit rates collapse against the score $\mathfrak{S}_{\delta,m,L} = \sqrt{\mathcal{I}_{m,L}(H)r_\delta(a, H)}/\tau(H, \zeta)$, matching the Gaussian approximation from Theorem 3.3.

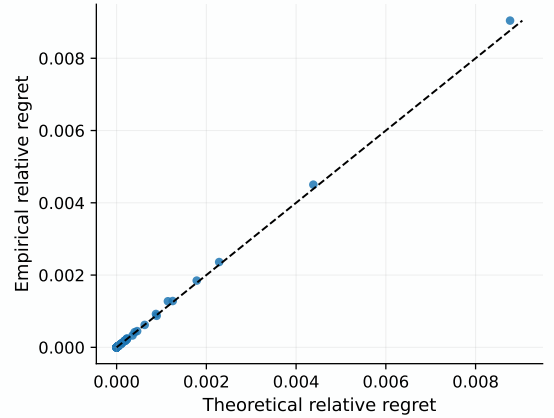


Figure 4: Quadratic operational regret. Across the tested grid, the empirical relative regret of the plug-in horizon aligns with the quadratic law from Proposition 3.5, showing that horizon mislocation costs are second order in log-scale error in the theorem regime.

Structural existence and operational identifiability need not coincide. The relevant scale is the score $\mathfrak{S}_{\delta,m,L}$.

4 Transfer Results and Operational Questions

The horizon law extends cleanly to finite projection families and sliced geometries under uniform projected control. Those results preserve the finite-sample versus staleness decomposition while changing the metric geometry and, therefore, the finite-sample exponent.

In fixed- ε Sinkhorn geometry, the pointwise and band-wise one-dimensional inheritance results are proved on compact support. On the embedded fixed-span model, the

moderate-band theorem is conditional on the centered self-coupling stability lemma and the associated uniform linearization control. The calibrated (8, 2) and (12, 2) regimes support that conditional picture, but they do not resolve it.

Operationally, online control begins with horizon inference: the plug-in horizon must first enter the useful-memory region and then remain there under streaming noise. The available evidence supports this mechanism on the tested profiles, but it does not yield a detector-uniform dominance law.

Finite-family transfer is theorem-level, the embedded Sinkhorn statement is conditional, and online control remains open.

5 Empirical Signatures

The experiments display the finite-sample signatures of the temporal-validity horizon, the useful-memory region induced by that horizon, and evidence relevant to the fixed- ε Sinkhorn frontier.

The Gaussian U-curve uses 20 Monte Carlo seeds, the roughness-family sweep uses 12, the fixed- ε Sinkhorn fits log-log slopes over sample sizes (24, 48, 96, 160) on the pairs (8, 1), (8, 2), (12, 1), and (12, 2) using 24 seeds, and the validity-detection experiment uses a piecewise-ramped drift stream with 12 seeds, a fixed operating window, and both ADWIN and Page-Hinkley detectors. The real-stream diagnostic uses the ELEC2 stream and a short anchor block on the marginal distribution of `nswprice`. The empirical goal is qualitative and structural: persistence of shape, ordering, and sign across repeated runs, not precise constant estimation for every curve.

Five signatures recur across the experiments. Finite-memory risk exhibits an interior optimum together with a near-optimal useful-memory band. That optimum moves with temporal roughness. The same shape appears on a real stream. The fixed- ε Sinkhorn frontier receives structured empirical support. Temporal validity can fail before changepoint evidence becomes statistically visible.

5.1 Interior Memory Optimum

The Gaussian horizon sweep isolates the variance-staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

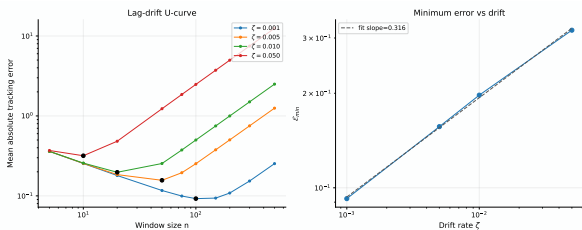


Figure 5: Interior risk optimum over memory length. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This interior optimum is the basic empirical signature of temporal validity under drift. Under stationarity, more

memory would continue to help. Under drift, the horizon that reduces variance also ages the evidence. Around the minimizing horizon, the curve displays a flat near-optimal region, which is the empirical analogue of the useful-memory region defined from the envelope Φ .

5.2 Roughness-Dependent Horizon Scaling

The roughness-family experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment traces the empirically optimal horizon across a sweep of ζ values.

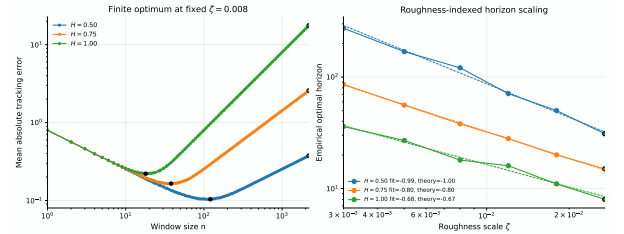


Figure 6: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H . The horizon is a function of drift regularity, not a fixed window rule.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.4.

5.3 Real-Stream Memory Diagnostic

A similar signature appears on a real nonstationary stream. Using ELEC2, consider the one-dimensional marginal law of `nswprice`. At each time, approximate the present target by a short anchor block of 48 observations and compare it, in W_1 , with the empirical law induced by the preceding memory window. Averaging this diagnostic across the stream yields a real-data analogue of the finite-memory U-curve.

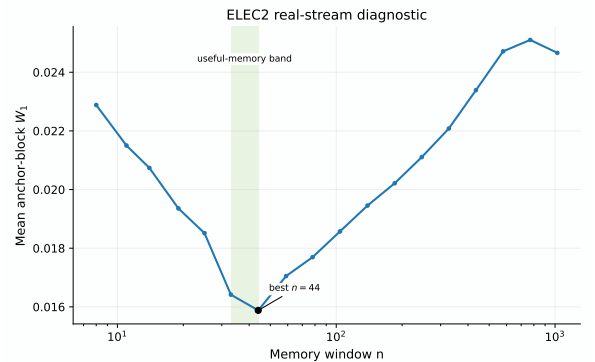


Figure 7: ELEC2 real-stream diagnostic on the `nswprice` marginal. The mean anchor-block W_1 error exhibits a finite interior optimum and a near-optimal useful-memory band, matching the temporal-validity signature on a real stream.

On ELEC2, the diagnostic optimum occurs at a finite memory scale, with a short near-optimal band around it.

The real stream therefore exhibits the same basic retention pattern as the synthetic horizon experiments: very short windows are too noisy, while very long windows become stale relative to the present block. On a robustness grid with anchor sizes 24, 48, and 96 and three contiguous segments of length 4000, the `nswprice` marginal retains an interior optimum in all nine tested configurations, with edge-to-best error ratios between 1.31 and 1.68. An interior pattern also appears on the `nswdemand` marginal in all nine tested configurations, although the optimizing scale moves more substantially across segments and anchor lengths; see Table 7.

5.4 Moderate-Band Sinkhorn Diagnostics

The fixed- ε Sinkhorn experiments address the bandwise question: when a practical geometry replaces raw W_2 , does the finite-sample term remain stable enough for the horizon law to stay informative in an embedded operational regime? The pointwise and bandwise one-dimensional fixed-support statements appear in propositions 2.19 and 2.20; the calibrated embedded fixed-span model extends the question to a broader low-dimensional setting.

The fixed- ε Sinkhorn geometry on embedded low-intrinsic-dimensional support keeps the finite-sample term close to the i.i.d. benchmark across a moderate regularization band, with no separate exponent emerging on the tested grid. In the calibrated fixed-span model with $\alpha = 2$ and span 0.25, the stable regime is widest on (8, 1), (8, 2), and (12, 2), and narrower on (12, 1). Direct Jacobian probes localize the remaining instability in the self-coupling blocks, while the calibrated certificate identifies a moderate-band regime on which the conditional theorem rests. Supporting tables and calibration details are kept in appendix B.

Intrinsic-dimension estimates from TwoNN track the stable-band cutoff better than ambient dimension on the tested grid, so the regime-selection variable is local geometry rather than ambient dimension.

5.5 Validity Loss Before Detection

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{val-det}} := t_{\text{detect}} - t_{\text{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the temporal-validity horizon, and t_{detect} is the first detector alarm that follows.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, and the qualitative lag persists under a different detector family as well. On the raw-observation input, the full tested grid of ADWIN and Page-Hinkley

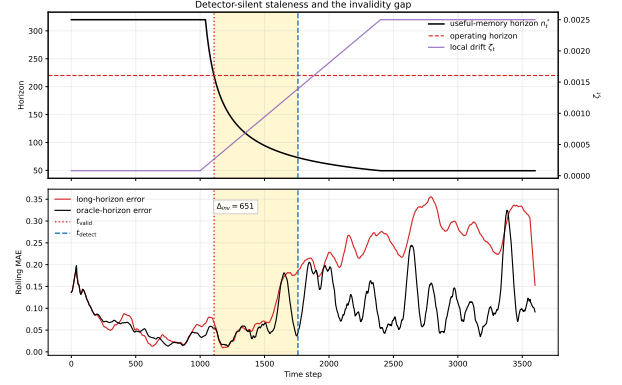


Figure 8: Temporal validity can fail before detector alarm. As drift intensifies, the temporal-validity horizon contracts until the operating horizon becomes stale at t_{valid} ; the detector responds only later at t_{detect} , leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

detectors, operating windows 160, 220, and 300, and the reported threshold sweep keeps positive-gap rate equal to one together with unit detection rate. Page-Hinkley reacts earlier on this drift family, with mean gaps between about 205 and 345, while ADWIN yields mean gaps between about 429 and 620. Residualized detector inputs are not uniformly better on this stream: ADWIN on absolute residuals fails to trigger on the tested grid, whereas Page-Hinkley still detects but with lower detection rates and substantially larger delays. The empirical claim is robustness of the sign of $\Delta_{\text{val-det}}$ on the tested observation-based detector/tuning family, not a detector-uniform delay law; see Table 8.

5.6 Empirical Summary

The experiments support a structural U-curve with a near-optimal useful-memory region, roughness-dependent horizon scaling, a corresponding pattern on the ELEC2 stream, structured support for the fixed- ε Sinkhorn frontier, and a measurable lag between validity loss and detector alarm.

6 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget analyses show that under drift, forgetting scales and window lengths must adapt to how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. The object studied here is earlier in the pipeline: it asks how long retained evidence remains temporally valid before age becomes the dominant source of tracking error.

Adaptive windowing and drift detection. Adaptive-window and concept-drift methods study when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Those works provide the reaction timescale. The horizon law adds the validity timescale: retained evidence can cease to represent the present before change statistics cross an alarm threshold.

Adaptive filtering and locally stationary smoothing. Classical adaptive filtering and state-space tracking tune forgetting factors, gain schedules, or process-noise levels to balance smoothing against responsiveness [Kalman, 1960, Jazwinski, 1970]. Locally stationary smoothing and bandwidth selection solve the analogous bias-variance problem for evolving regression or spectral targets [Dahlhaus, 1997, Vogt, 2012]. These literatures supply mechanisms for adaptation. The horizon law provides the distributional law that such mechanisms must respect when the retained object is an empirical distribution.

Transport geometry and operational calibration. Wasserstein distance and Sinkhorn-type estimators provide the finite-sample geometries that enter the horizon law [Fournier and Guillin, 2015, Boissard and Le Gouic, 2014, Weed and Bach, 2019, Genevay et al., 2019, Mena and Niles-Weed, 2019, del Barrio et al., 2023, Goldfeld et al., 2024]. Sliced and projection-robust variants show that geometry choice can change the finite-sample exponent, while intrinsic dimension and regularization calibration determine which regime is available [Kolouri et al., 2018, Deshpande et al., 2019, Paty and Cuturi, 2019, Lin et al., 2020, Nadjahi et al., 2020, Nietert et al., 2022, Genans and Wintenberger, 2026, Facco et al., 2017].

Age and nonstationarity. Wasserstein nonstationarity measures and Age-of-Information formulations make age into an operational variable [Jiang et al., 2020, Keehan et al., 2025, Amini and Vinas, 2026, Kaul et al., 2012, Yates et al., 2021]. The temporal-validity horizon gives that variable a distributional interpretation: age is costly because it accumulates geometric mismatch with the present.

Lag-geometry inference. Lag-geometry inference under noisy transport observations is complementary to the structural law. Parreira [2026b] study power-law lag geometry through a parameter-coupled heteroscedastic regression, deriving an information law for estimating (H, ζ) and testing the lag model itself.

7 Limitations and Open Problems

The theorem-level results comprise the temporal-validity horizon, the induced useful-memory region, the exact uniform-window staleness bound, a tractable one-dimensional root- n proof model, a structural Gaussian lower bound at the exponent level, a Gaussian location benchmark, and inferable-horizon coverage and regret under lag-geometry input. Other results concern narrower transfer, operational, or empirical statements.

Scope of the closed theory. The fully proved finite-sample instantiation is the one-dimensional root- n regime under bounded support and fixed span. The structural lower bound is subclass based, and the exact Gaussian constants are internal to the available proof scheme rather than class-tight across broader deterministic Hölder path classes. Finite-family projection transfer is theorem-level under uniform projected control, but richer dominated

families and raw multivariate Wasserstein lie beyond the proved range.

Operational frontier. For fixed- ε Sinkhorn, the compact-support one-dimensional pointwise and bandwise inheritance results are explicit, while the embedded low-dimensional moderate-band theorem remains conditional on the centered self-coupling stability lemma and the associated uniform linearization control. The strongest evidence is concentrated on moderate bands bounded away from zero, especially on the calibrated $(8, 2)$ and $(12, 2)$ regimes.

Empirical scope. The experiments target structural signatures: U-curves, roughness scaling, real-stream retention, and validity-detection separation. They do not identify detector-uniform delay laws or competitive end-to-end performance.

Open problems. The remaining frontier is organized by four questions.

- (1) **Embedded Sinkhorn closure.** Prove the centered self-coupling stability lemma and the associated uniform linearization bound needed to turn the conditional moderate-band embedded fixed- ε statement into a theorem.
- (2) **Multivariate transfer beyond finite families.** Extend theorem-level transfer from finite projection families to richer dominated multivariate classes, and determine how far this route can go toward raw multivariate Wasserstein tracking.
- (3) **Lower theory beyond the Gaussian benchmark.** Sharpen the lower theory beyond the current Gaussian location benchmark, both for broader deterministic Hölder path classes and for finite-sample regimes beyond root- n .
- (4) **Memory design and online control.** Determine when non-uniform weights enlarge the useful-memory region beyond constant factors, derive stable online estimators of (H, ζ) under lag noise, and obtain high-probability guarantees that a data-driven controller remains inside the useful-memory region.

8 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes finite memory a temporal-validity object with a finite horizon.

The horizon law

$$\mathbb{E} d(\hat{P}_t^{(n)}, P_t) \leq C_K n^{-a} + C_S \zeta n^H,$$

and its optimizer

$$n^*(a, H) \asymp (C_K / \zeta)^{1/(a+H)}$$

summarize that balance. This law induces a useful-memory region around the optimal scale.

The core theory closes in a tractable one-dimensional root- n proof model, matched with a structural Gaussian

lower bound and a Gaussian location benchmark. In fixed- ε Sinkhorn geometry, the one-dimensional inheritance results place the horizon law in an operational transport geometry.

The empirical signatures align with the theory: finite-memory U-curves, roughness-dependent horizon scaling, a real-stream finite-memory pattern, and a positive lag between validity loss and detector alarm. Horizon inference adds a second distinction: a temporal-validity horizon can exist structurally and still be operationally hard to estimate from lag geometry alone.

The open frontier consists of embedded Sinkhorn closure, non-uniform memory design, and online control.

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Proof Details

This appendix gives the detailed derivations for structural results whose full calculations would interrupt the main line.

A.1 Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^h r^{2H} \quad \text{and} \quad G_h := \sum_{r=1}^h (h^H - (h-r)^H)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\text{ramp}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{S_h}\right)$$

and

$$L_h^{\text{min}}(\sigma, \zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma} \sqrt{G_h}\right).$$

Proof of proposition 2.11. Fix $H \in (0, 1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma} \sqrt{S_h} \rightarrow \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\text{ramp}}(\sigma, \zeta)}{\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}} \rightarrow A^H \Phi\left(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\right).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0, \infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h \geq 1} L_h^{\text{ramp}}(\sigma, \zeta) \sim C_H^{\text{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\text{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \geq g_h - (h-r)^H = h^H - (h-r)^H$$

for every feasible endpoint-saturating profile g . Therefore the profile

$$g_r^{\text{min}} = h^H - (h-r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^h g_r^2$. Its rescaled energy satisfies the Riemann-sum limit. Thus

$$h^{-(2H+1)} G_h \rightarrow I_H,$$

where

$$I_H = \int_0^1 (1 - (1-u)^H)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

With the same scaling

$$h = A \left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma} \sqrt{G_h} \rightarrow A^{H+1/2} \sqrt{I_H}.$$

Setting $x = A^{H+1/2} \sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h \geq 1} L_h^{\min}(\sigma, \zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\text{ramp}}} = \left(\frac{I_H^{-1}}{2H+1} \right)^{H/(2H+1)} = \left(\frac{H+1}{2H^2} \right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$.

A.2 Two-point lower bound used in proposition 2.10

The structural lower bound above is the exponent-level consequence of the exact Gaussian two-point calculation. For the ramp pair

$$\mu_{t-j}^{\pm, h} = \pm \zeta(h-j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\text{ramp}}(\sigma, \zeta)$. Minimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H+1)^{-1}$ for the root- n regime.

B Finite-Sample and Operational Diagnostics

This appendix gives finite-sample diagnostics for the theorem line and the operational frontier. These diagnostics support theorem-facing regimes without changing claim status.

B.1 Finite-Sample Geometry Diagnostic

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log-log slope in n . We also probe the fixed-span one-dimensional triangular window directly by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that diagnostic the within-window span ζn^H is held fixed. Finally, we probe the Sinkhorn diagnostic at a few fixed values of ε .

In low dimension, the raw i.i.d. Wasserstein term is compatible with a root- n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array diagnostic, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in the 1D proof-model proposition.

Table 2: Empirical scaling diagnostic for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array diagnostic compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn diagnostic is reported at fixed ε . These rows are numerical support for the operational finite-sample discussion rather than theorem statements.

Setting	Estimated slope	Comment
$W_2, d = 1$	-0.48	close to $-1/2$
$W_2, d = 2$	-0.41	slower decay
$W_2, d = 4$	-0.25	slower decay
$W_2, d = 5$	-0.24	slower decay
W_2 , i.i.d. mixture, $d = 1, H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2 , i.i.d. mixture, $d = 1, H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2 , triangular, $d = 1, H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	stability constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	stability constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	stability constant changes

B.2 Bahadur Remainder Diagnostics

Two deformation families make the proof-model remainder assumption concrete. The first is a smooth fixed-support monotone warp class $g_\delta(x) = x + \delta x(1-x)$ with $|\delta| \leq \Delta < 1$. On the tested spans 0.2 and 0.4, the integrated remainder decays at rates between 1.40 and 1.54, and the scaled quantity $n \int_0^1 R_n(u)^2 du$ still decays at rates between 0.40 and 0.54. The second is the fixed-span triangular-array uniform-kernel model used in the proof line. There the integrated residual rate is about 1.43 for the triangular sample and 1.49 for the i.i.d. mixture benchmark, while the residual-to-MSE ratio drops from about 0.38 at $n = 25$ to about 0.11 at $n = 400$.

These diagnostics locate the role of drift span explicitly. Within the tested fixed-span range, the remainder stays lower-order and the empirical rate remains well above the n^{-1} threshold needed for item (iv). Rougher kernels and growing-span experiments remain less regular, which is why the analysis keeps the general triangular-array verification problem open.

B.3 Extended and Operational Regimes

The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed- ε Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 3: Representative rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed- ε Sinkhorn across ε on embedded low-intrinsic-dimensional support.

Regime	Setting	Parameter	Rate exponent α
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.4652
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.10	0.10	0.5722
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.4737
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.25	0.25	0.5588
extended	triangular ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.4792
extended	iid mixture ambient $d=8$, intrinsic $k=1$, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.50$	0.50	0.5599
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.50$	0.50	0.5706
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.20$	0.20	0.5466
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.20$	0.20	0.6150
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.10$	0.10	0.4646
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.10$	0.10	0.5060
operational	Sinkhorn iid mixture ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.05$	0.05	0.4339
operational	Sinkhorn triangular ambient $d=8$, intrinsic $k=1$, $\varepsilon=0.05$	0.05	0.5124

The extended-regime rows indicate that low intrinsic dimension preserves a rate near the root- n regime with-

out a dramatic triangular-versus-benchmark gap. The operational-regime rows record empirical evidence relevant to the fixed- ε Sinkhorn setting in definition 2.16: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they align with the embedded band-wise statement. A full proof of triangular-array inheritance would close this loop.

Two complementary diagnostics sharpen the same boundary. First, direct probes of the centered Sinkhorn scaling Jacobian on the embedded fixed-span model show that bands approaching zero are the numerically unstable region, whereas on a moderate band such as $[0.2, 0.8]$ the cross-coupling block remains contractive and the self-coupling inverse norms appear to plateau with sample size. Second, on the supported synthetic frontier grid, intrinsic-dimension estimates predict the observed stable-band cut-off better than ambient dimension. These diagnostics identify a useful regime statistic and isolate the band where the instability is concentrated. They do not upgrade the embedded fixed- ε statement from conjecture to theorem.

The calibrated summary separates three ingredients on the same embedded grid: the empirically stable band, the theorem-ready band obtained from the structural certificate, and the self-coupling proxy on the calibrated pairs where that proxy is currently informative. On $(8, 2)$ and $(12, 2)$, the empirical and theorem-ready bands both reach 0.5, and the two self-coupling blocks keep positive spectral gap, worst spectral radius below 0.97, and largest-sample mean inverse norm below 3.0.

Table 4: Moderate-band Sinkhorn summary on the calibrated embedded grid. The table reports the empirical stable band, the theorem-ready band, the minimum recovered finite-sample rate exponents inside the theorem-ready band, and whether both self-coupling blocks satisfy the spectral-radius and inverse-norm proxy thresholds.

Pair	Emp. $\varepsilon_{\max}$	Thm.-ready $\varepsilon_{\max}$	Min iid a	Min tri. a	Self
(8, 1)	0.5	0.5	0.488	0.458	—
(8, 2)	0.5	0.5	0.455	0.419	yes
(12, 1)	0.2	0.2	0.501	0.474	—
(12, 2)	0.5	0.5	0.428	0.493	yes

Table 5: Calibrated moderate-band Sinkhorn certificates on the embedded fixed-span grid. The table reports the largest regularization value for which the certificate returns a certified band on the tested synthetic frontier.

Ambient / intrinsic	α	Max certified ε
(8, 1)	2.0	0.5
(8, 2)	2.0	0.5
(12, 1)	2.0	0.2
(12, 2)	2.0	0.5

B.4 Smooth Multivariate Transfer Diagnostics

The multivariate transfer diagnostic uses a drifting three-dimensional mean vector and a smooth log-sum-exp functional. The finite-sample error of the window estimator, the drift-induced bias of the window target, and the linearization residual are tracked separately across window lengths. The target behavior is the same as in the abstract multivariate transfer lemma: the finite-sample piece

decreases with window length, the drift piece increases, and the linearization residual remains lower order than the finite-sample term on the tested windows.

Window	Tot.	Fin.	Drift	Resid.
8	0.2009	0.1988	0.0165	0.0239
16	0.1513	0.1542	0.0355	0.0162
32	0.1373	0.1363	0.0735	0.0066
64	0.1492	0.0807	0.1503	0.0051

Table 6: Smooth multivariate transfer diagnostic under triangular-array drift. The finite-sample term decreases with window length, the drift term increases, the total error has an interior minimum, and the linearization residual stays lower order across the tested windows.

B.5 Robustness Summaries

The real-stream, validity-gap, and online-controller sweeps are summarized here in compact form. They delimit where the qualitative signatures remain stable on the tested grids.

Variable	Runs	Interior	Best-window range	Edge/best range
nswprice	9	9/9	14–435	1.31–1.68
nswdemand	9	9/9	25–327	1.15–3.24

Table 7: ELEC2 robustness summary across anchor sizes 24, 48, and 96 and segment starts 0, 8000, and 16000, with segment length 4000. The interior-optimum count records how often the best window stays away from the search boundary; the edge/best range reports the ratio between the worse boundary error and the optimum.

Detector input	Detector	Detection rate	Mean-gap range	Excess-area range
Observation	ADWIN	1.00	429–620	7.41–20.41
Observation	Page-Hinkley	1.00	205–345	3.55–8.20
Abs. residual	ADWIN	0.00	—	—
Abs. residual	Page-Hinkley	0.33–0.92	994–1665	16.94–151.24

Table 8: Validity-detection robustness summary across operating windows 160, 220, and 300 and the reported threshold grid. Mean-gap and excess-area ranges are computed over the successful runs in each detector/input family.

Profile	Seed	Plug-in/oracle	Activity/oracle	Structural/oracle	Adaptive/oracle
Default	0	4.14	1.41	1.33	1.30
Default	1	4.17	1.33	1.38	1.31
Rough	0	2.50	1.65	1.34	1.49
Rough	1	2.18	1.40	1.26	1.30

Table 9: Reduced online-controller sweep with profiles **default** and **rough**, seeds 0 and 1, and maximum one-step window-index jumps. Entries report mean absolute error relative to the oracle controller.

References

- O. Deniz Akyildiz, Pierre Del Moral, and Joaquín Míguez. Gaussian entropic optimal transport: Schrödinger bridges and the sinkhorn algorithm. *CoRR*, abs/2412.18432, 2024. doi: 10.48550/arXiv.2412.18432.
- Arash A. Amini and Luciano Vinas. Wasserstein concentration of empirical measures for dependent data via the method of moments. *CoRR*, abs/2601.07228, 2026. doi: 10.48550/arXiv.2601.07228.

- R. R. Bahadur. A note on quantiles in large samples. *The Annals of Mathematical Statistics*, 37(3):577–580, 1966.
- Omar Besbes, Yonatan Gur, and Assaf Zeevi. Non-stationary stochastic optimization. *Operations Research*, 63(5):1227–1244, 2015. doi: 10.1287/OPRE.2015.1408.
- Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007. doi: 10.1137/1.9781611972771.42.
- Emmanuel Boissard and Thibaut Le Gouic. On the mean speed of convergence of empirical and occupation measures in wasserstein distance. *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques*, 50(2):539–563, 2014. doi: 10.1214/12-AIHP517.
- Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087. PMLR, 2019. URL <https://proceedings.mlr.press/v89/cheung19b.html>.
- Rainer Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 1997. doi: 10.1214/aos/1034276620.
- Eustasio del Barrio, Alberto Gonzalez-Sanz, Jean-Michel Loubes, and Jonathan Niles-Weed. An improved central limit theorem and fast convergence rates for entropic transportation costs. *SIAM Journal on Mathematics of Data Science*, 5(3):639–669, 2023. doi: 10.1137/22M149260X.
- Ishan Deshpande, Yuan-Ting Hu, Ruoyu Sun, Ayis Pyrros, Nasir Siddiqui, Sanmi Koyejo, Zhizhen Zhao, David Forsyth, and Alexander G. Schwing. Max-sliced wasserstein distance and its use for GANs. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10648–10656, 2019. doi: 10.1109/CVPR.2019.01090.
- Stephan Eckstein and Marcel Nutz. Quantitative stability of regularized optimal transport and convergence of sinkhorn’s algorithm. *SIAM Journal on Mathematical Analysis*, 54(6):5922–5948, 2022. doi: 10.1137/21M145505X.
- Elena Facco, Maria d’Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1):12140, 2017. doi: 10.1038/s41598-017-11873-y. URL <https://doi.org/10.1038/s41598-017-11873-y>.
- Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162(3–4):707–738, 2015. doi: 10.1007/s00440-014-0583-7.
- João Gama, Pedro Medas, Gladys Castillo, and Pedro Rodrigues. Learning with drift detection. In *Advances in Artificial Intelligence — SBIA 2004*, volume 3171 of *Lecture Notes in Computer Science*, pages 286–295. Springer, 2004. doi: 10.1007/978-3-540-28645-5_29.
- João Gama, Indrè Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):44:1–44:37, 2014. doi: 10.1145/2523813.
- Ferdinand Genans and Olivier Wintenberger. Geometry-aware optimal transport: Fast intrinsic dimension and wasserstein distance estimation. *CoRR*, abs/2602.04335, 2026. doi: 10.48550/arXiv.2602.04335.
- Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583, 2019.
- Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.
- Michel Groppe and Shayan Hundrieser. Lower complexity adaptation for empirical entropic optimal transport. *Journal of Machine Learning Research*, 25(344):1–55, 2024. URL <https://jmlr.org/papers/v25/23-0671.html>.
- Andrew H. Jazwinski. *Stochastic Processes and Filtering Theory*, volume 64 of *Mathematics in Science and Engineering*. Academic Press, New York, 1970.
- Jiashuo Jiang, Xiaocheng Li, and Jiawei Zhang. Online stochastic optimization with wasserstein based non-stationarity, 2020. URL <https://arxiv.org/abs/2012.06961>.
- Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960. doi: 10.1115/1.3662552.
- Sanjit Kaul, Roy Yates, and Marco Gruteser. Real-time status: How often should one update? In *Proceedings of IEEE INFOCOM*, pages 2731–2735, 2012. doi: 10.1109/INFOCOM.2012.6195689.
- Dominic S. T. Keehan, Edward J. Anderson, and Wolfram Wiesemann. Don’t look back in anger: Wasserstein distributionally robust optimization with nonstationary data. *CoRR*, abs/2510.18566, 2025. doi: 10.48550/arXiv.2510.18566.
- Soheil Kolouri, Gustavo K. Rohde, and Heiko Hoffmann. Sliced wasserstein distance for learning gaussian mixture models. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3427–3436, 2018. doi: 10.1109/CVPR.2018.00361.
- Tianyi Lin, Chenyou Fan, Nhat Ho, Marco Cuturi, and Michael I. Jordan. Projection robust wasserstein distance and riemannian optimization. *Advances in Neural Information Processing Systems*, 33:9383–9397, 2020.

- Alessio Mazzetto and Eli Upfal. Nonparametric density estimation under distribution drift. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24251–24270. PMLR, 2023. URL <https://proceedings.mlr.press/v202/mazzetto23a.html>.
- Gonzalo Mena and Jonathan Niles-Weed. Statistical bounds for entropic optimal transport: Sample complexity and the central limit theorem. *Advances in Neural Information Processing Systems*, 32:4543–4553, 2019.
- Kimia Nadjahi, Alain Durmus, Lenaic Chizat, Soheil Kolouri, Shahin Shahrampour, and Umut Simsekli. Statistical and topological properties of sliced probability divergences. *Advances in Neural Information Processing Systems*, 33:2085–2095, 2020.
- Sloan Nietert, Ziv Goldfeld, Ritwik Sadhu, and Kengo Kato. Statistical, robustness, and computational guarantees for sliced wasserstein distances. *Advances in Neural Information Processing Systems*, 35:28179–28193, 2022.
- Vinicius Parreira. Power-law lag geometry: Structural inference from noisy transport estimates, 2026b. Preprint, July 2026.
- François-Pierre Paty and Marco Cuturi. Subspace robust wasserstein distances. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 5072–5081, 2019.
- Austin J. Stromme. Minimum intrinsic dimension scaling for entropic optimal transport. *CoRR*, abs/2306.03398, 2023. doi: 10.48550/arXiv.2306.03398.
- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998.
- Michael Vogt. Nonparametric regression for locally stationary time series. Technical Report CWP22/12, cemmap, 2012. URL <http://www.cemmap.ac.uk/wps/cwp221212.pdf>.
- Zhiyong Wang, Jize Xie, Yi Chen, John C. S. Lui, and Dongruo Zhou. Variance-dependent regret bounds for non-stationary linear bandits, 2024. URL <https://arxiv.org/abs/2403.10732>.
- Jonathan Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A):2620–2648, 2019. doi: 10.3150/18-BEJ1065.
- Roy D. Yates, Yin Sun, D. Richard Brown, Sanjit K. Kaul, Eytan Modiano, and Sennur Ulukus. Age of information: An introduction and survey. *IEEE Journal on Selected Areas in Communications*, 39(5):1183–1210, 2021. doi: 10.1109/JSAC.2021.3064980.
- Jianjun Yuan and Andrew G. Lamperski. Trading-off static and dynamic regret in online least-squares and beyond. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence*, pages 6712–6719. AAAI Press, 2020. doi: 10.1609/AAAI.V34I04.6149.